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Power of real gurus

PHYSICS

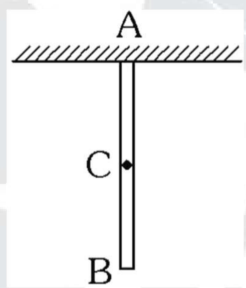
TARGET : JEE-ADVANCED-2024

CAPS-2^{+}

1. There is an isolated planet having mass 2M and radius 2R, where M and R are the mass and radius of the earth. A simple pendulum having mass m and length 2R is made to small oscillations on the planet. the time period of SHM of pendulum is 3384 K seconds. Find the value of K. (take π = 3.00, g = 10 m/s^2, $\sqrt{2} = 1.41$)

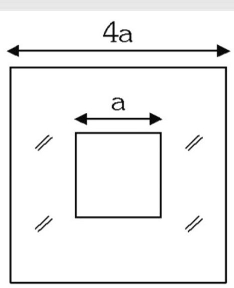
(A) 1 (B) 2 (C) 3 (D) 4

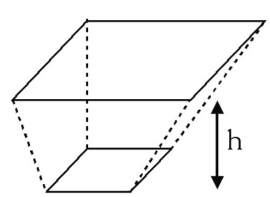
2. A wire of uniform cross section is hanging vertically and due to its down weight its length changes. There is a point 'C' on the wire such that change in length AC is equal to the change in length BC points A, B & C are shown in figure. Find AC/BC. Take $\left(\sqrt{2} = 1.41\right)$



(A) 0.21 (B) 0.41 (C) 0.61 (D) 0.81

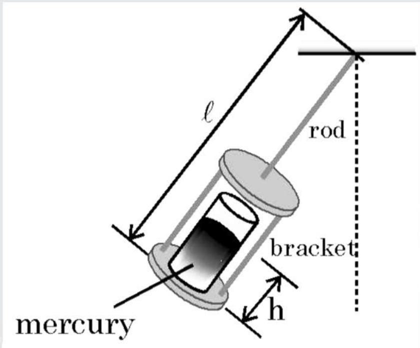
3. Figure shows a soap film formed between two square figures made of a uniform wire. The bigger square is held while keeping it in a horizontal plane and the smaller square is slowly allowed to drop vertically. It reaches an equilibrium state after dropping a height h. Let surface tension of soap = T. Mass per unit length of wire = l Acceleration due to gravity = g. Given that h = $h = \dfrac{n\lambda g a}{2\sqrt{4T^2 - \lambda^2 g^2}}$; find the integer value n.





(A) 3 (B) 4 (C) 2 (D) 1

4. In order to compensate for deviations caused by temperature changes, a pendulum clock uses a large cylindricalglass tube filled with mercury (volumetric coefficient of expansion $\beta_{mercury}$) as a pendulum bob. This tube is held by a rod and bracket made of brass (linear coefficient of expansion α_{brass}) as shown in figure. The combined length of the rod and bracket is ℓ (measured from the point of suspension of the pendulum). Neglecting the mass of the brass and the glass and neglecting the expansion of the glass, find the height of the mercury in the glass tube must be so that the centre of mass of the mercury is to remain at a fixed distance from the point of suspension, regardless of temperature.



Pendulum with a temperature compensator

(A) $h = \left(\dfrac{4\alpha_{brass}}{3\beta_{mercury}}\right)$ (B) $h = \left(\dfrac{3\alpha_{brass}}{\beta_{mercury}}\right)\ell$ (C) $h = \left(\dfrac{3\alpha_{brass}}{2\beta_{mercury}}\right)\ell$ (D) $h = \left(\dfrac{2\alpha_{brass}}{\beta_{mercury}}\right)\ell$

5. A hypothetical spherical planet of radius R, and its density varies as ρ= Kr, where K is constantand r is the distance from the center. The pressure caused by gravitational pull inside (r < R)the planet at a distance r measured from its center given by [Assume pressure on the surfaceof planet is P_0]

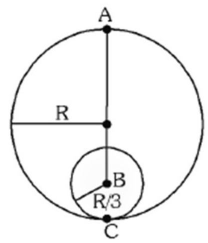
(A) $P_0 + \frac{1}{4}K^2G\pi(R^4 - r^4)$

(B) $P_0 + \frac{2}{27}K^2G\pi(R^4 - r^4)$

(C) $P_0 + \frac{K^2G\pi}{4}(r^4)$

(D) None of these

6. Inside a uniform sphere of mass M and radius R, a cavity of radius R/3 is made in the sphere as shown.



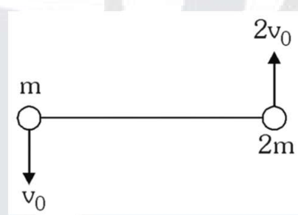
(A) Gravitational Field inside the cavity is uniform.

(B) Gravitational field inside the cavity is non-uniform.

(C) The escape velocity of a particle of very small mass projected frompoint A is $\sqrt{\frac{88GM}{45R}}$

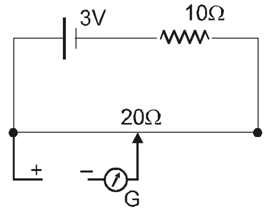
(D) The escape velocity of a particle of very small mass projected fromany point on the surface of the sphere is $\sqrt{\frac{88GM}{45R}}$

7. Two point masses of mass m and 2m placed on smooth horizontal plane are connected by a light rod of length ℓ. Masses are given velocities in horizontal plane perpendicular to rod as shown. If area of cross section of rod is A and Young's modulus is Y, then elongation in the rod, will be



(A) $\frac{mv_0^2}{AY}$ (B) $\frac{2mv_0^2}{AY}$ (C) $\frac{6mv_0^2}{AY}$ (D) $\frac{3mv_0^2}{AY}$

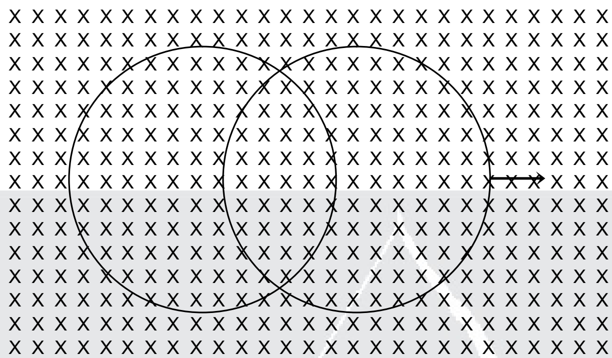
8. A 10 m long potentiometer wire has a resistance of 20 ohm. It is connected in series with a battery of emf 3V and a resistance of 10 Ω. The internal resistance of cell is negligible. If the length can be read accurately up to 1 mm, the potentiometer can read voltage.



(A) up to minimum of 0.2 mV (B) with an accuracy of 0.2 mV

(C) with an accuracy of 0.1 mV (D) up to maximum of 2V

9. In the diagram shown, a uniform magnetic field is present perpendicular to the plane of the paper. Both the rings are identical and have a constant resistance per unit length. The left ring has been kept fixed at its position and the right ring is slid uniformly on the left ring towards the right hand side. Which of the following statements is/are true ?



(A) The emf induced in the left ring is zero

(B) The emf induced in both the rings is non zero..

(C) The magnetic force acting on the right ring is zero.

(D) The magnetic force acting on both the rings is non-zero.

10. A particle of mass 'm' moves under a conservative force with potential energy $U(x) = \frac{cx}{x^2 + a^2}$ where c and a are positive constants. Choose the correct option(s)

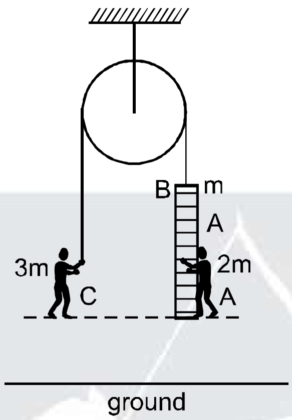
(A) Position of unstable equilibrium is x = +a

(B) The particle can oscillate about x = –a

(C) If particle's speed is more than $\sqrt{\frac{c}{ma}}$ at origin, then it will reach $x = +\infty$

(D) If particle's speed in more than $\sqrt{\frac{c}{ma}}$ at x = –a, then it will reach $x = -\infty$

11. Man A of mass 2m is standing on a ladder of mass m. Ladder is attached with an inextensible light string and man C of mass 3m is holding the other end of string. Initially, the whole system is at rest. Assume whole system is at sufficient height and pulley is frictionless. Also A and C are at same horizontal level. Then choose the correct options. [$g = 10\ m/s^2$].



(A) If man A starts climbing up on the ladder with a constant acceleration of 4 m/s^2 w.r.t ground then, the vertical distance between A and C after 1 sec is 1 m.

(B) If man C starts climbing up on the rope (while A remain at rest with respect to ladder) with constant acceleration of 4 m/s^2 with respect to ground then, the vertical distance between A and C after 1 sec is 2 m.

(C) If both man A and C starts climbing up with constant acceleration 4 m/s^2 with respect to ground then, acceleration of ladder is 4 m/s^2 upward .

(D) If both man A and C starts climbing up with constant acceleration 4 m/s^2 with respect to ground then, acceleration of ladder is 3 m/s^2 upward .

12. A body is thrown from ground with an angle α returns to ground after covering a horizontal distance b. Then this body is projected with same velocity at same angle α above horizontal from a tower of height h_1 and returns to the ground after covering a horizontal distance s. If this body is projected with same speed at angle α (below the horizontal) from tower of height h_2 so that it will fall on the ground at the same distance s from this tower base, then Choose the correct option(s) :

(A) $h_1 + h_2 = \frac{2s^2}{b}\tan\alpha$

(B) $h_2 - h_1 = 25\cot\alpha$

(C) $\frac{h_2}{h_1} = \frac{s+b}{s-b}$

(D) $\frac{h_2}{h_1} = \frac{s}{2b}$

Paragraph-1 for Q. No. (13 & 14)

In our solar system, asteroids, small satellites, comets with diameters less than 600 km can be very irregular in shape, whereas those with large diameters are spherical. Only if the rocks have sufficient strength to resists gravity then an object can maintain a non-spherical shape. In solar system, there is a planetary object of uniform density ρ and radius R.

13. Find the compressive stress S (defined as force per unit cross-sectional area) near the centre of the planetary object

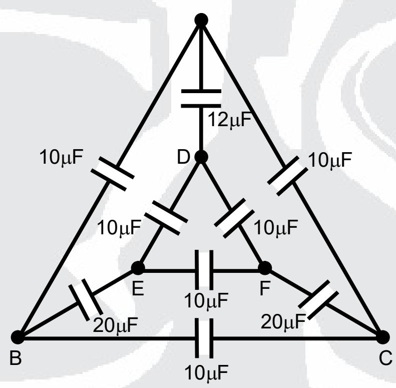
(A) $S = \pi G\rho^2R^2$ (B) $S = \frac{2}{3}\pi G\rho^2R^2$ (C) $S = \frac{1}{3}\pi G\rho^2R^2$ (D) $S = \frac{4}{3}G\rho^2\pi R^2$

14. What is the largest possible size of a7 non-spherical self gravitating satellite made of concrete? Assume that concrete has maximum compressive stress of 4.0×10^7 N/m^2 and a density $\rho = 3000$ kg/m^3.

(A) 250 km (B) 140 km (C) 180 km (D) 640 km

Answer Q.15 Q.16 and Q.17 by appropriately matching the information given in the three columns of the following table.

In the figure the nine capacitors are connected as shown.



Column-I (Charge drawn from battery)	**Column-II** (Charge in capacitor connected between A & D)	**Column-III** (Charge in capacitor connected between B & C)
(I) 100 μC	(i) 120 μC	(P) Zero
(II) 210 μC	(ii) Zero	(Q) 40 μC
(III) 200 μC	(iii) 40 μC	(R) 100 μC

15. If a 10 V battery is connected across the terminal A and D

(A) (I) (ii) (P) (B) (III) (i) (P) (C) (II) (i) (Q) (D) (III) (iii) (R)

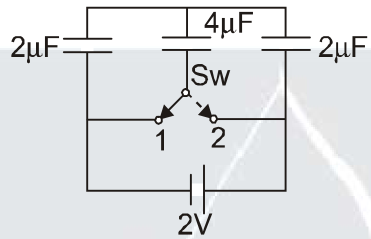
16. If a 10 V battery is connected across the terminal B and C

(A) (I) (ii) (P) (B) (II) (iii) (R) (C) (II) (ii) (R) (D) (III) (iii) (R)

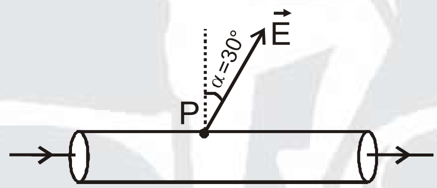
17. If a 10 V battery is connected across the terminal E and F

(A) (I) (i) (P) (B) (II) (ii) (Q) (C) (III) (ii) (Q) (D) (III) (iii) (R)

18. What amount of heat will be generated in the circuit shown in the figure, after the switch S*w* is shifted from position 1 to position 2?



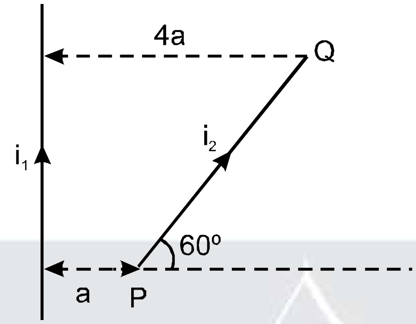
19. There is a long solid conductor of resistivity $\rho = 1 \times 10^{-6}$ Ω-m, surrounded by air. There is a steady electric current along the length of the conductor. At point 'P' just outside the cylinder the electric field strength E = 10^{-4} V/m is directed at an angle of α to the normal to the surface. If the current density in the conductor in the vicinity of point 'P' (see fig.) is 10a, find a.



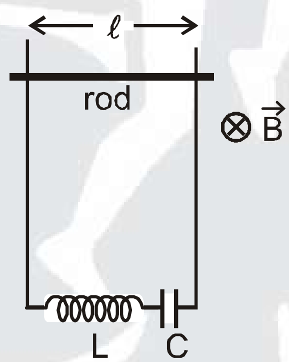
20. To measure the diameter of a wire, a screwgauge is used. In a complete rotation, spindle of the screw gauge advances by $\frac{1}{2}$ mm and its circular scale has 50 divisions. The main scale is graduated to $\frac{1}{2}$ mm. If the wire is put between the jaws, 8 main scale divisions are clearly visible and 10 divisions of circular scale co–inside with the reference line. The resistance of the wire is measured to be (10Ω ± 1.5%). Length of the wire is measured to be 10 cm using a scale of least count 1mm. Maximum percentage error in resistivity measurement is : (in nearest integer)

21. Consider a long solenoid of radius R which has n turns per unit length. A time dependent current $I = I_0 \sin \omega t$ flows in solenoid, magnitude of electric field at a perpendicular distance r < R, from the axis of symmetry of solenoid is found to be E = $\frac{\alpha}{4}\omega\mu_0 nI_0 r \cos\omega t$, where α is pure number and μ_0 is permability of free space. Find α. :

22. A long straight conductor carries current 'i_1'. A wire PQ carrying current 'i_2' is placed as shown. The net force on PQ is $\frac{2\mu_0 i_1 i_2}{\pi}\ell nx$, then write the value of 'x'.



23. Figure shows a conducting horizontal rod of resistance r is made to oscillate simple harmonically with a fixed amplitude in a uniform and constant magnetic field B, directed inwards. The ends of rod always touch two parallel fixed vertical conducting rails. The ends of rails are joined by an inductor and a capacitor having self inductance and capacitance $\frac{1}{\pi}$ Henry and $\frac{1}{\pi}$ farad respectively. The amplitude of current in the circuit depends on the frequency of oscillation of rod. Find the time period of rod in seconds for the amplitude of the current to be maximum ? (do not consider self inductance anywhere other than in the inductor)



24. A man of mass 50 kg falls freely from a height of 40 m into a swimming pool and just come to rest at the bottom of the pool. Assume that the average upward force on the man due to water is 1000 N. Find the depth of water in the pool in meter. (g = 10 m/s^2)

25. Two blocks of masses m_1 = 10 kg and m_2 = 20 kg are connected by a spring of stiffness k = 200 N/m. The coefficient of friction between the blocks and the fixed horizontal surface is μ = 0.1. Find the minimum constant horizontal force F (in Newton) to be applied to m_1 in order to slide the mass m_2. (Take g = 10 m/s^2)

ANSWER KEY

1.	(B)	**2.**	(B)	**3.**	(A)	**4.**	(D)	**5.**	(A)
6.	(AC)	**7.**	(C)	**8.**	(BD)	**9.**	(AD)	**10.**	(ABCD)
11.	(AC)	**12.**	(AC)	**13.**	(B)	**14.**	(C)	**15.**	(B)
16.	(C)	**17.**	(B)	**18.**	$Q = 4\ \mu J$	**19.**	5	**20.**	3
21.	2	**22.**	$x = 2$	**23.**	2	**24.**	40	**25.**	20